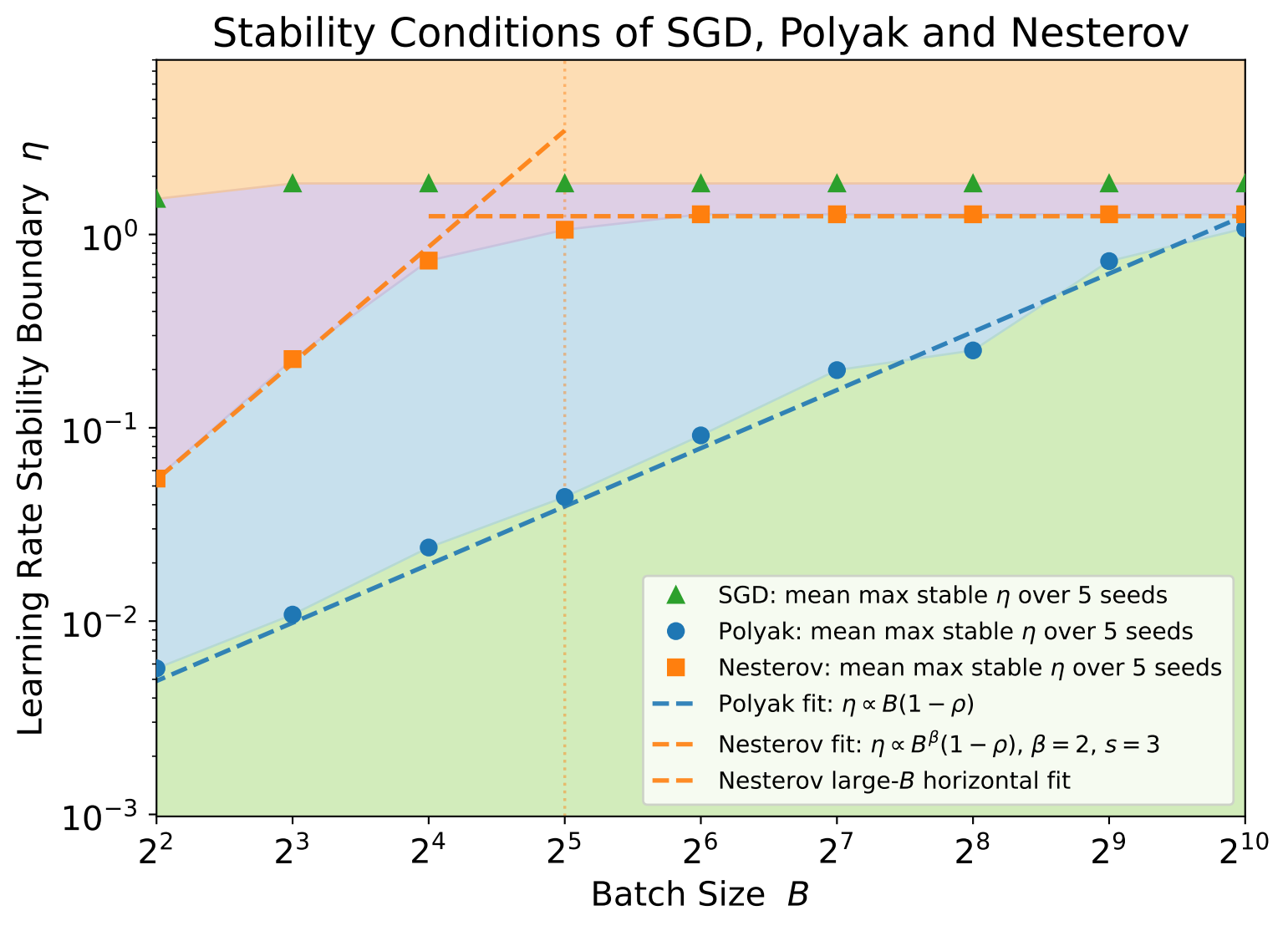
Stability Conditions of SGD, Polyak and Nesterov
Learning Rate Stability Boundary η
10^{0}
10^{-1}
10^{-2}
10^{-3}
2^{2}
2^{3}
2^{4}
2^{5}
2^{6}
2^{7}
2^{8}
2^{9}
2^{10}
Batch Size B
SGD: mean max stable η over 5 seeds
Polyak: mean max stable η over 5 seeds
Nesterov: mean max stable η over 5 seeds
Polyak fit: $\eta \propto B(1-\rho)$
Nesterov fit: $\eta \propto B^{\beta}(1-\rho)$, $\beta = 2$, $s = 3$
Nesterov large-B horizontal fit